The three laws of thermodynamics 8a

The term thermodynamics comes from the Greek language; thermo has to do with heat, a form of energy, and dynamics refers to movement. So thermodynamics is about how heat energy moves. Three important laws of thermodynamics were developed in the mid 19th century and the early 20th century.

The first law of thermodynamics: Conservation of energy

Energy changes from one form to another, it cannot be created or destroyed. Energy is always conserved. The total amount of energy in the universe stays constant.

When a bar heater gives off heat and also some light, these forms of energy have come from another energy form: electric energy. Neither the heat nor the light were created.

Einstein's famous equation $E = mc^2$ broadened this law. The total amount of energy and matter in the universe stays the same. Therefore, people are able to change some matter into huge amounts of energy, like in a nuclear reactor. In this case, the energy did not come from another form of energy, it came from matter. The law of conservation of energy is only true when the amount of matter stays the same, which is most of the time.

The second law of thermodynamics: Entropy

The word **entropy** means messiness or disorder. Energy systems increase their entropy; that is, they become more and more messy. Look at your bedroom – after a few days, it can look really untidy. Its entropy has increased. If everything in your room ends up on the floor, the potential energy (here gravitational potential energy) has become less. Over time there is more entropy – that is, more chaos, more disorder, more mess – less orderliness and less potential energy.

Furthermore, heat travels from hotter objects to colder ones in order to make everything equally warm. Then it is no longer an ordered system, where heat would be 'over here' and cold would be 'over there'.

The third law of thermodynamics: Absolute zero

All particles of matter have stopped moving when they reach a temperature called absolute zero. Absolute zero is 0 K (Kelvin) or -273°C. At 0 K, gases have practically no volume.

Answer the comprehension questions, writing full sentences which contain the question as well as the answer.

On the lines

1 What language does the term **thermodynamics** come from?

_____

2 What is moving in the laws of thermodynamics?

_____

3 Does the amount of energy in the universe increase or decrease?

_____

4 Does the amount of entropy in the universe increase or decrease?

_____

5 What is all matter made of?

_____

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Between the lines

6 How old is the earliest law?

7 How old is the youngest law?

8 What is the difference between the law of conservation of energy and the famous equation from Einstein?

9 What does the term **conservation** mean?

10 Does your room have more entropy or energy in the morning? What about the evening?

11 Which is colder: 0 K or 0°C?

Beyond the lines

12 Think of two other terms that contain 'therm'.

13 Suggest why so many scientific terms are based on Greek or Latin words.

14 List five forms of energy.

15 What is the unit of measurement for energy?

16 What is Einstein's famous equation, and what do the symbols in it stand for?

17 What has to happen to increase order in your bedroom?

18 What is the boiling point of water in degrees Celsius and in Kelvin?

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